

Supporting Information for ”Primary and net ecosystem production in a large lake diagnosed from high-resolution oxygen measurements”

Bieito Fernández Castro^{1,2}, Hannah Elisa Chmiel¹, Camille Minaudo¹,

Shubham Krishna¹, Pascal Perolo³, Serena Rasconi⁴, Alfred Wüest^{1,5}

¹Physics of Aquatic Systems Laboratory, Margaretha Kamprad Chair, Institute of Environmental Engineering, École Polytechnique

Fédérale de Lausanne, Lausanne, Switzerland.

²Ocean and Earth Science, National Oceanography Centre, University of Southampton, SO14 3ZH, Southampton, UK

³Institute of Earth Surface Dynamics, University of Lausanne, 1015 Lausanne, Switzerland

⁴Université de Savoie Mont-Blanc, INRAE, CARRTEL, 74200 Thonon les Bains, France

⁵Eawag, Swiss Federal Institute of Aquatic Science and Technology, Surface Waters – Research and Management, Kastanienbaum,

Switzerland.

Contents of this file

1. Text S1 to S2

2. Figures S1 to S8

Introduction This file contains two sections of supplementary text describing the implementation of the diel methods in the spectral and time domain, respectively, as well as 7 figures.

Text S1. Implementation of the diel spectral method

The diel spectral method used to calculate GPP rates is based on the notion that the amplitude of the diel DO signal ($A_{\text{DO},24\text{h}}$, considered from the zero crossing to the crest of a sinusoidal-like signal) can be related to hourly nighttime respiration rate as:

$$r_{\text{night}} \approx \frac{2}{t_{\text{night}}} A_{\text{DO},24\text{h}} [\text{mmol m}^{-3} \text{h}^{-1}] \quad (1)$$

where t_{night} is the night duration in hours. Assuming that daytime respiration rate is equal to the nighttime rate as in the classical diel method (Staehr et al., 2010), the total daily respiration (R is defined positive) is:

$$R = 24 \cdot r_{\text{night}} [\text{mmol m}^{-3} \text{d}^{-1}] \quad (2)$$

The daily GPP rate is then simply, $\text{GPP} = \text{NEP} + R$. The underlying assumption is that NEP is treated as a fortnight time-scale metabolic balance, and the NEP fluctuations at higher frequencies are neglected.

By assuming a sinusoidal form, the amplitude of the diel signal ($A_{\text{DO},24\text{h}}$) can be related to the variance ($\sigma_{\text{DO},24\text{h}}^2$) calculated by integration of the DO power spectrum (S_{DO}) around a period of 24 h (in practice we used periods between 18 and 28 hours):

$$A_{\text{DO},24\text{h}} = (2 \cdot \sigma_{\text{DO},24\text{h}}^2)^{1/2} \quad (3)$$

Before integration, the background physical signal of processes occurring at these scales was removed in four steps (see Fig. S1): (Step i) In a first step, the fraction of the DO variance correlated with temperature variations (which results from vertical displacements due to baroclinic motions) was removed using the co-spectrum of both variables, $S_{\text{DO-T}}$ (Levine & Lueck, 1999). A transfer function, $H_{\text{DO-T}} = S_{\text{DO-T}}/S_{\text{T}}$ was calculated, where S_{T} is the temperature spectrum, and used to calculate a “clean” DO spectrum as $\hat{S}_{\text{DO}} = S_{\text{DO}} - |H|^2 S_{\text{T}}$. (Step ii) Even in the clean spectrum, a non-zero variance baseline was

observed at frequencies around the diel peak, meaning that some physical signal remains. A baseline for the oxygen signal around the diel peak was calculated by fitting the spectra to a straight line in logarithmic space, $S_n = a \cdot f^b$, where f is frequency for the frequency range between 12 and 48 hours excluding the diel peak (18–28 hours, see Fig. S1). The diel DO variance, $\sigma_{\text{DO},24\text{h}}^2$ was then calculated as:

$$\sigma_{\text{DO},24\text{h}}^2 = \sigma_{\hat{S},24\text{h}}^2 - \sigma_{S_n,24\text{h}}^2 \quad (4)$$

where $\sigma_{\hat{S},24\text{h}}^2$ and $\sigma_{S_n,24\text{h}}^2$ are the integrals of $\hat{S}$ and S_n in the frequency range corresponding to periods between 18 and 28 hours:

$$\sigma_{S,24\text{h}}^2 = \int_{1/28\text{h}}^{1/18\text{h}} S(f) df \quad (5)$$

(Step iii) In order to further remove spurious signals, $\sigma_{\text{DO},24\text{h}}^2$ smaller than 1.5 times $\sigma_{S_n,24\text{h}}^2$ were set to zero. (Step iv) Finally, based on the fact that part of the oxygen variation is directly linked to photosynthesis and depends on light, we remove any remaining anomalous signals. This was done by calculating the phase difference ($\delta_{\text{DO-light}}$) between the DO and incident shortwave radiation signals using the co-spectrum of the two variables. Segments in which the oxygen increase preceded the light increase (i.e. an oxygen increase was observed before sunrise, equivalent to negative phase difference) were considered as non-biological. In order to account for variability and errors in the spectral calculations, we discarded the segments where the phase difference was $\delta_{\text{DO-light}} < -2$ hours and, for those segments, we also set $\sigma_{\text{DO},24\text{h}}^2$ to zero.

The spectral calculations were done at each time-step (1 hour) over 256-datapoint (~ 11 days) overlapping segments of DO fluctuations with respect to the low-passed signal ($\text{DO}' = \text{DO} - \langle \text{DO} \rangle$). The segments were subdivided in 128-datapoint half-overlapping

sub-segments for power spectral calculations using the Welch method (Welch, 1967). The sensors sampling within the mixed layer might be affected by the air-lake DO flux, which has a diel component (Fig. 1.D). In order to account for this, the DO' record within the mixed-layer for each segment was corrected prior to the spectral calculations by subtraction of the time-integral of $F^{\text{air-lake}}$:

$$DO''(t) = DO'(t) - \frac{1}{h_{\text{ML}}} \int_{t_0}^{t_1} F^{\text{air-lake}}(t) dt \quad (6)$$

where h_{ML} is the mixed-layer depth, and t_0 and t_1 are the initial and final times of the record segment.

Text S2. Implementation of the diel method in the time domain

In order to obtain daily values of GPP, R and NEP in the time-domain, 24 hours segments of hourly *nep* rates (Eq. 11 in the main manuscript) were fitted to the following equation (Hanson et al., 2008; Obrador et al., 2014):

$$nep^i = \underbrace{P_{max} \tanh\left(\frac{\alpha I_0 e^{-k_d z^i}}{P_{max}}\right)}_{gpp^i} - \underbrace{R_{20} \theta^{(T^i - 20)}}_{r^i} \quad (7)$$

where P_{max} is the maximum photosynthetic rate at saturating light ($\text{mmol m}^{-3} \text{ h}^{-1}$), α is the photosynthetic efficiency ($\text{mmol m}^{-2} \text{ h}^{-1} \text{ W}^{-1} \text{ m}^2$), I_0 is the surface incident light from the meteorological station, k_d is the diffuse attenuation coefficient derived from PAR measurements, R_{20} is the respiration rate at 20°C ($\text{mmol m}^{-3} \text{ h}^{-1}$), and $\theta \approx 1.07$ is a coefficient accounting for thermal dependence of respiration. Daily GPP, R and NEP were then calculated as the daily means of the model-derived rates.

The coefficient of determination R^2 of fit of DO vs T anomalies provides a quantitative measure of the fraction of daily DO variance that is attributable to physical processes. Whenever this variance exceeded 75% ($R^2 > 0.75$) we considered that the physical signal

was too dominant for the diel biological cycle to be detected with enough confidence. Furthermore, during some days, anomalous negative GPP or R were computed, as a result, for example, of a nighttime increase in DO concentration. In all these cases, we set $GPP = NEP$ ($R = 0$) when $NEP > 0$ and $R = |NEP|$ ($GPP = 0$) when $NEP < 0$, where NEP was calculated by averaging Eq. 11 in the main manuscript over 24 h, without using the inverse model results.

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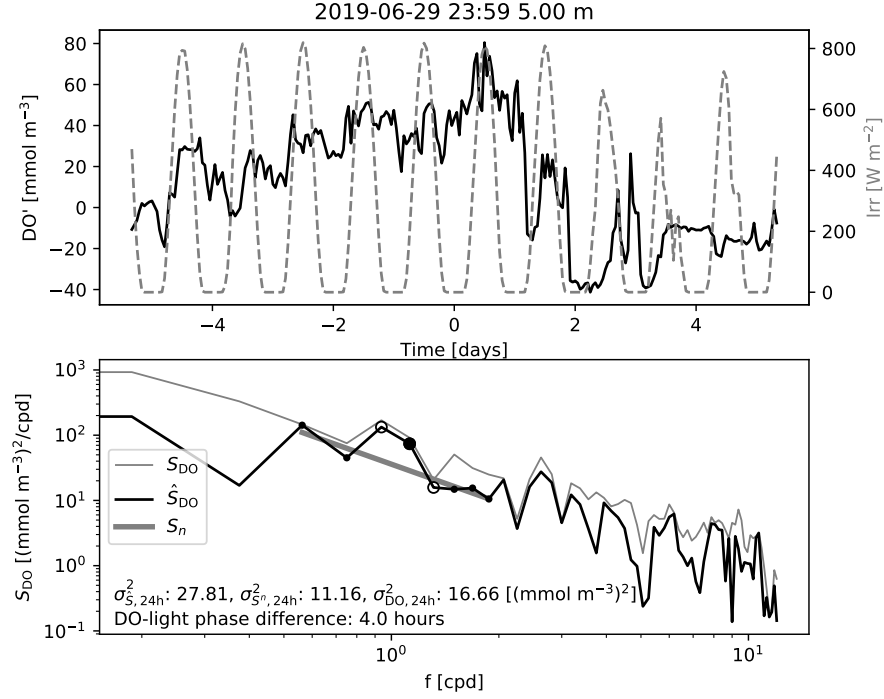


Figure S1. Example of time-series of DO anomaly ($DO' = DO - \langle DO \rangle$, where $\langle DO \rangle$ is the low-passed DO signal) and irradiance (upper panel), and DO spectra for a 10-day data segment centered at midnight on 29 June. The thin gray line in the lower panel represents the raw DO spectrum (S_{DO}), the black line represents the DO spectrum de-noised using the spectral correlation with the temperature signal ($\hat{S}_{DO}$). The baseline spectrum (S_n) is shown as a gray thick line. Small black dots in the $\hat{S}$ were used to fit S_n . Big dots correspond to frequencies in the range $1/28 \text{ h} < f < 1/18 \text{ h}$, used to calculate the variance around the diel period.

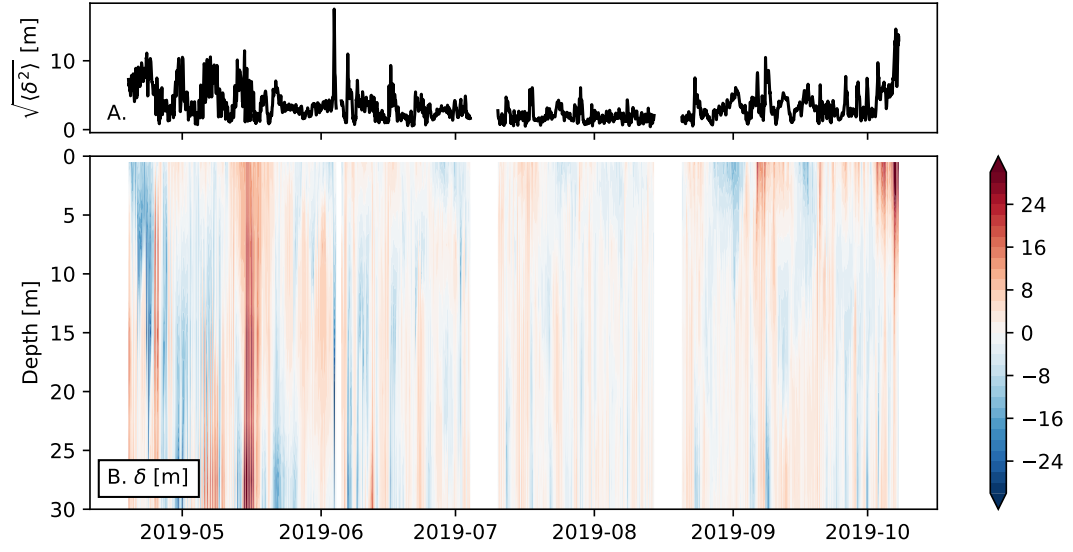


Figure S2. Isothermal displacements estimated from temperature anomalies ($T' = T - \langle T \rangle$) with respect to the low-passed temperature ($\langle T \rangle$) according to Eq. 2 in the main manuscript. Panel A shows root mean square displacements.

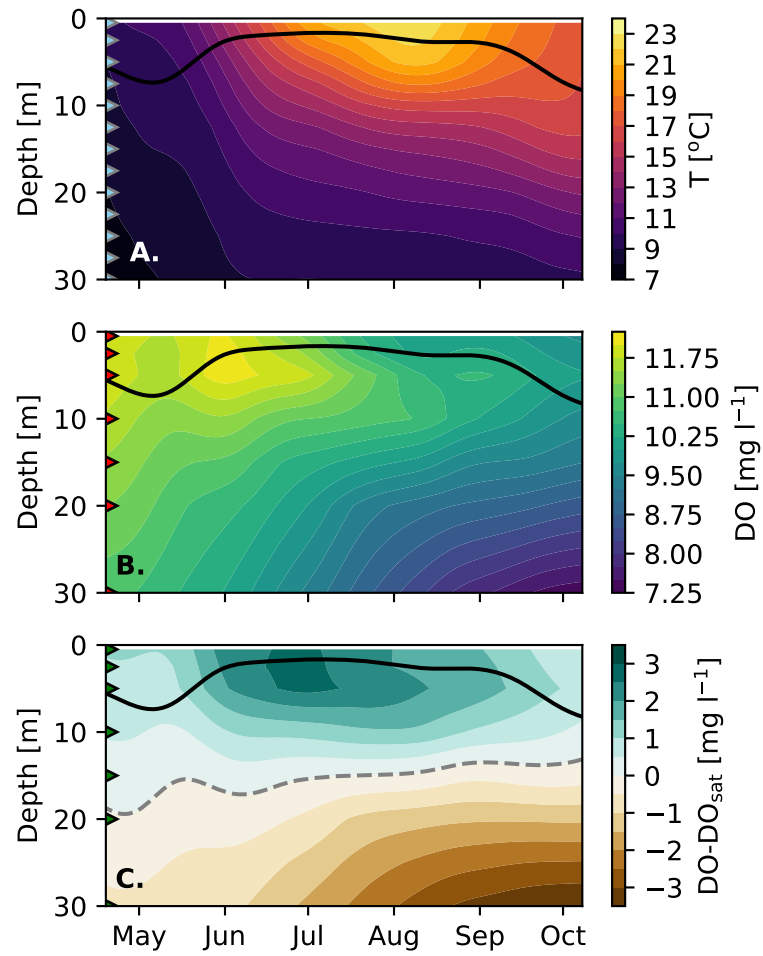


Figure S3. Smoothed time-series of A. temperature, B. dissolved oxygen and C. oxygen anomaly with respect to saturation concentration. The solid black line represents the mixed layer depth, defined as the depth where temperature exceeds the value of the upper sensor (0.5 m) by 1 °C, and the gray dashed line, the zero saturation isoline. The depth of the sensors is indicated by right-pointing triangles.

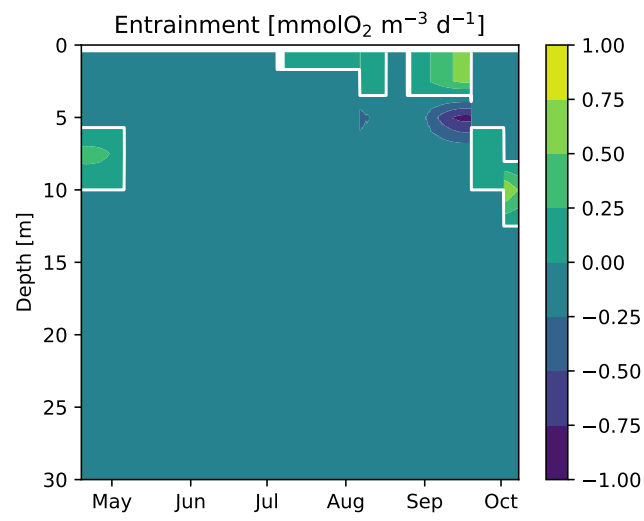


Figure S4. Divergence of the mixed-layer entrainment flux in the seasonal oxygen budget.

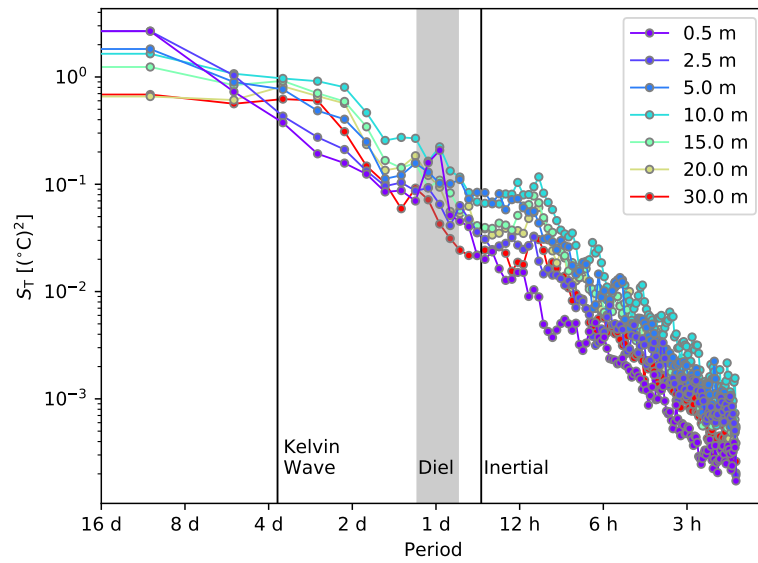


Figure S5. Temperature power spectra for the sensors co-located with oxygen sensors. The periods of the Kelvin and inertial waves are indicated as vertical lines. The shaded area indicates periods around the diel cycle (18-28 hours).

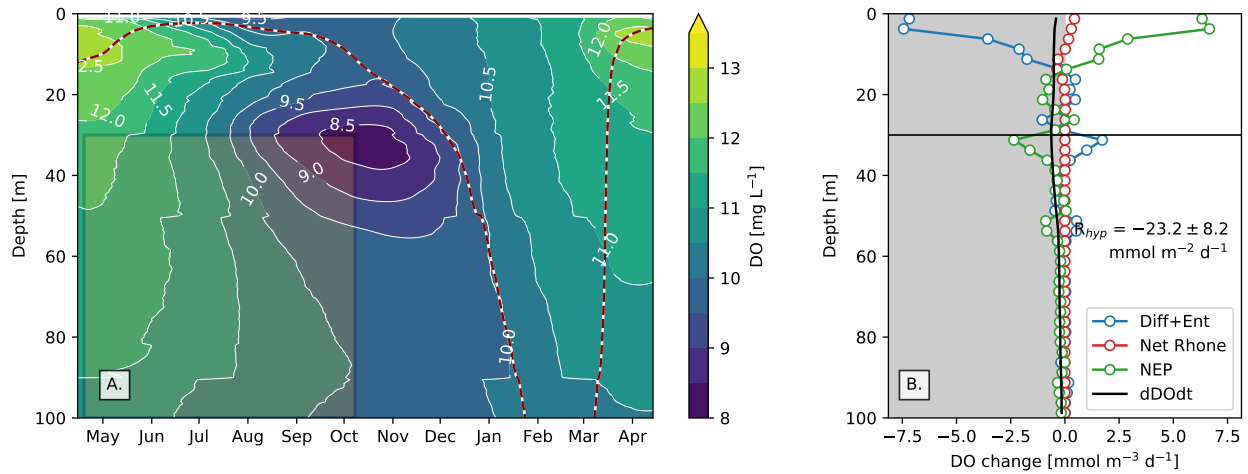


Figure S6. Hypolimnetic oxygen budget computed using DO profiles collected with a CTD. A. 15-day smoothed DO time-series, B. mean profiles of the budget terms between 19 April and 7 October 2019. Depth-integrated hypolimnetic respiration for the depth interval 30–100 m and for the same time period is reported. In panel A, the mixed layer is displayed as a dotted line and the integration period as a shaded rectangle.

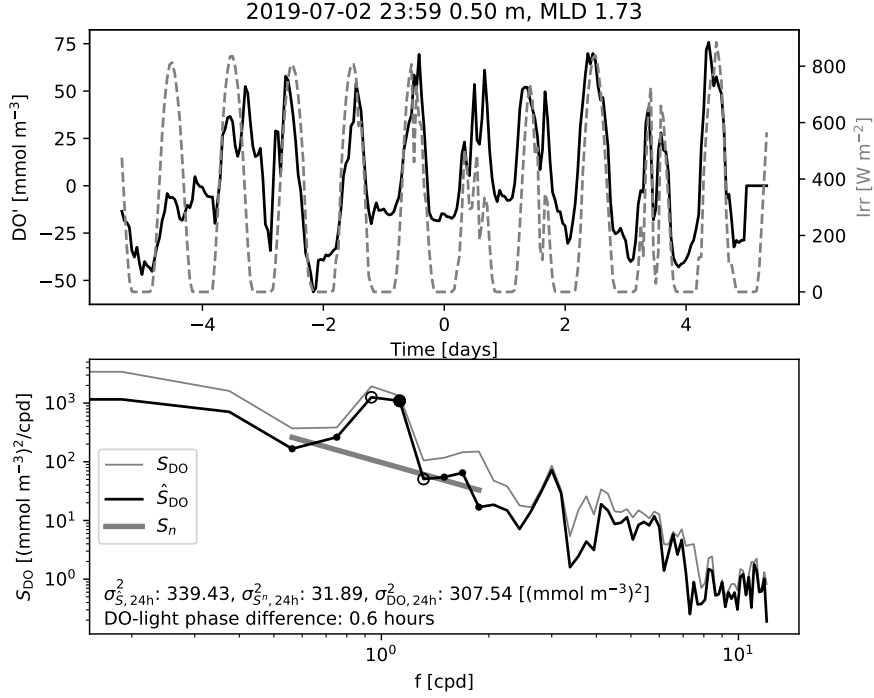


Figure S7. Example of time-series of DO anomaly ($DO' = DO - \langle DO \rangle$, where $\langle DO \rangle$ is the low-passed DO signal) and irradiance (upper panel), and DO spectra for a 10-day data segment centered at midnight on 2 July collected with the upper sensor at 0.5 m depth. The thin gray line represents the raw DO spectrum (S_{DO}), the black line represents the DO spectrum de-noised using the spectral correlation with the temperature signal ($\hat{S}_{DO}$). The baseline spectrum (S_n) is shown as a gray thick line. Small black dots in the $\hat{S}$ were used to fit S_n . Big dots correspond to frequencies in the range $1/28 \text{ h} < f < 1/18 \text{ h}$, used to calculate the variance around the diel period.

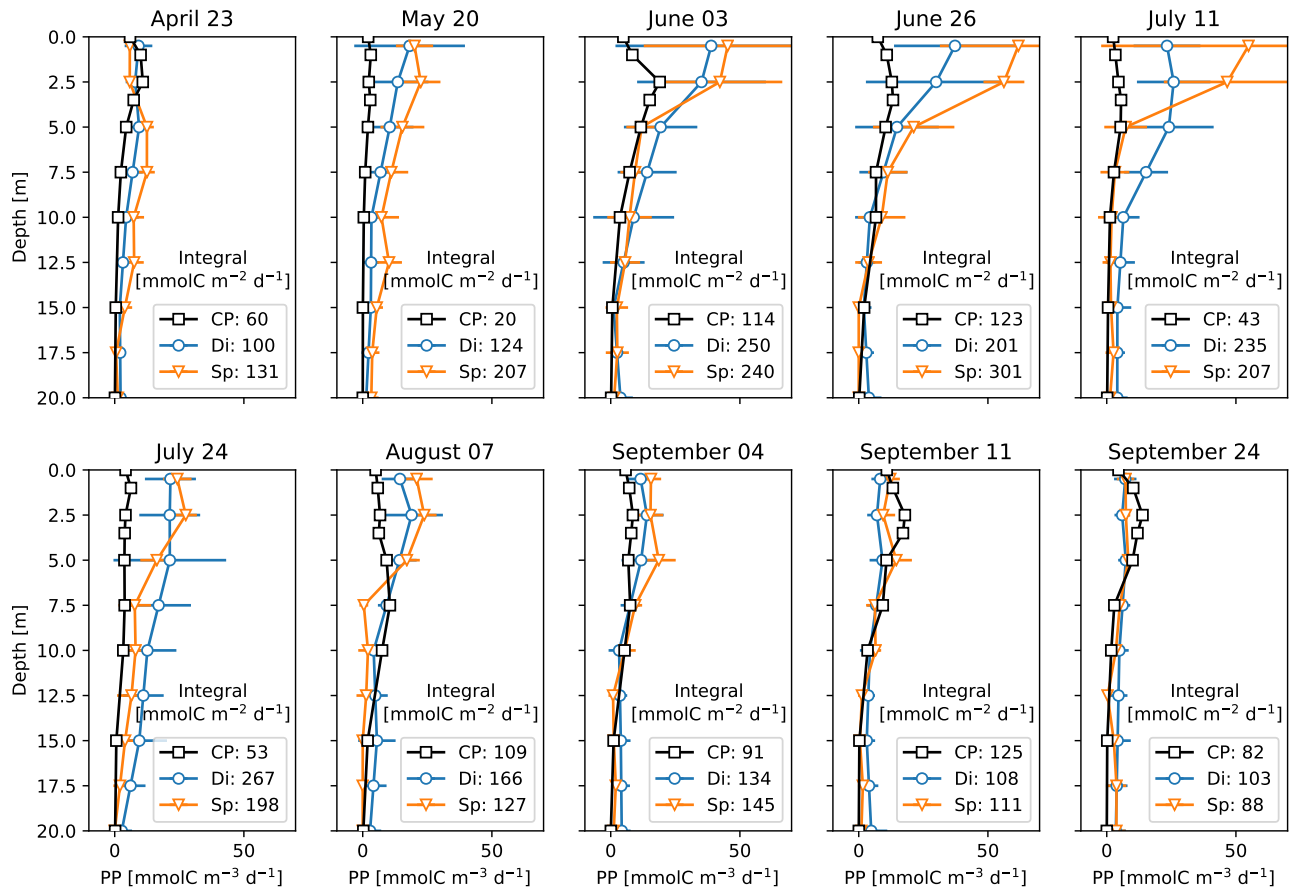


Figure S8. Primary production profiles from the ^{13}C method at the SHL2 site during the 2019 productive period (black) together with weekly means of GPP estimates derived from in situ DO measurements using the time-domain diel method (blue) and the scale separation spectral method (orange). Standard deviation of the weekly values are shown as horizontal lines. Depth-integrated values are also displayed. All the rates were converted to carbon units using a oxygen to carbon Redfield molar ratio of 138:106.